3 **(a)** A commonly used quantity in astronomy is *luminosity*. The luminosity of a star is the total energy radiated by the star per second.

(i) The luminosity of our sun is 3.826×10^{26} J s^{-1}. The mean distance of the Earth from the sun is 1.496×10^{8} km. Determine the intensity of light reaching the Earth.

intensity = …….. W m^{-2} [2]

(ii) A student, using a photometer that measures the intensity of visible light, measures the intensity of sunlight at noon to be less than the value calculated in **(a)(i)**. Suggest a reason for this observation.

……..……..

……..

…….. [1]

(iii) A photometer of area 4.00×10^{2} cm^{2} is aimed directly at the sun at the top of a building. Determine the maximum power incident on the photometer.

maximum power = ……..…….….. W [2]

(b) Fig. 3.1 shows the displacement y of a particle in a sinusoidal wave as a function of time t.

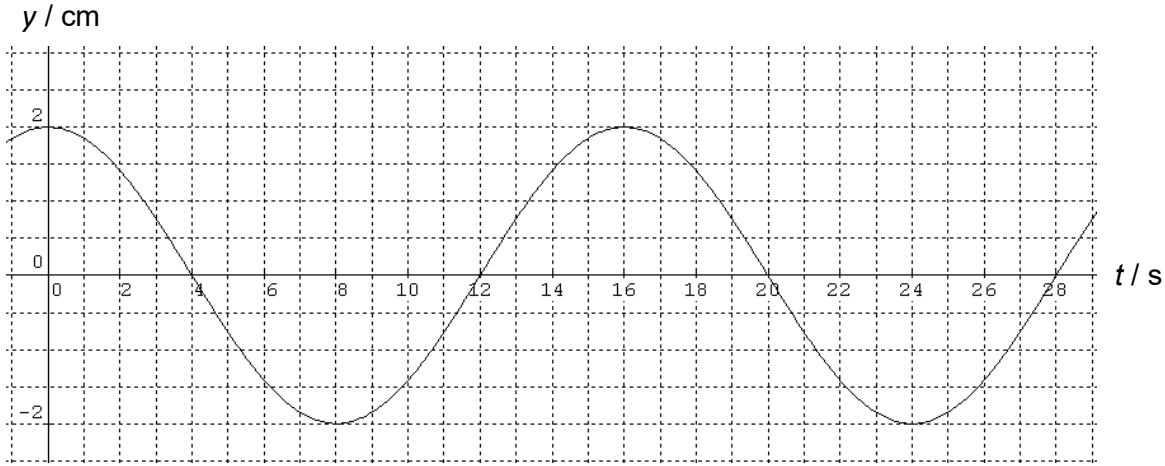

Fig 3.1

(i) Using Fig. 3.1, write an equation that represents the displacement y of the particle in terms of t.

y = ... [2]

(ii) A second particle is situated nearer to the source of the wave, at a distance $\frac{\lambda}{4}$ from the first. Determine the phase difference between the vibrations of the two particles.

phase difference = ... [2]

(iii) Sketch in Fig. 3.1 to illustrate the variation with time of the displacement of the second particle. [2]